

ENTRY 056 Large-Circuit Tier Doubled (12 → 24) — August 25, 2026 — Status: COMPLETE

Entry 056 — Doubling the 60+ Physical Qubit Tier and Widening Capacity to 90/130

Deepening the Underrepresented Large-Circuit Slice, Per Updated Project Direction

1. Motivation

Following Entry 055's dataset-growth pause, focus shifted to the large-circuit tier: only 12 star-GHZ circuits at $k=10/12/14$ existed (from Entry 052), a thin slice relative to the 3,205-circuit bell-pair-dominated set. `entry056_grow_bigger.py` topped up `PER_SIZE` from 2 to 4 per size per chip using the same resumable `have_per_size` logic as Entry 052, adding 12 new circuits (doubling the tier to 24) and pushing the observed maximum from 65 to 67 physical nodes.

2. A New Padding Ceiling Was Hit and Fixed Before Training

The new circuits pushed max edges to 106, exceeding the Entry 053/054/055 padding ceiling of `MAX_E=104`. This was caught before training rather than silently truncating data: `MAX_N/MAX_E` were widened to 90/130 (headroom above the new 67/106 observed max, same margin convention as every prior capacity bump).

3. Results: Held or Improved Everywhere

metric	Entry 055 (n=3,205)	Entry 056 (n=3,217)
same-chip MAE	1.16 pts	1.13 pts
same-chip R²	0.970	0.976
Kyiv→Sherbrooke MAE / R² / fc	4.08 / 0.786 / 8.52	4.10 / 0.814 / 8.60
Sherbrooke→Kyiv MAE / R² / fc	2.87 / 0.894 / 3.72	2.65 / 0.905 / 2.87

Adding 12 large, structurally different circuits (deep routed star topologies vs. mostly short bell pairs) did not destabilize the model at the wider 90/130 capacity -- same-chip accuracy improved slightly, Kyiv-to-Sherbrooke R^2 improved (0.786 to 0.814) with MAE and floor-collapse essentially flat, and Sherbrooke-to-Kyiv improved on every metric (MAE 2.87 to 2.65, R^2 0.894 to 0.905, floor-collapse MAE 3.72 to 2.87 -- a new best across all entries in this direction).

Entry 056 — Conclusion

Growing the large-circuit tier and widening capacity a second time (80/104 to 90/130) composed cleanly with every prior fix, matching the pattern already seen across Entries 053-055: capacity growth and dataset growth are largely independent axes that do not trade off against each other in this architecture.

4. Next Steps

Uncertainty quantification is scoped next (Entry 057) as the primary research-value addition, run alongside continued large-circuit and dataset growth as capacity allows.

Research Log — Index Addendum (Entry 056)

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Scripts: `entry056_grow_bigger.py`, `entry056_build_graphs.py`, `entry056_train_gnn.py`, `entry056_cross_chip.py`. Data: `quantumbridge_data/entry052_biggerghz_dataset.json` (24 records), `entry056_graph_dataset.json` (3,217 total), `entry056_gnn_results.json`, `entry056_cross_chip_results.json`. Max circuit size grew 65 to 67 nodes; `MAX_N/MAX_E` widened 80/104 to 90/130.